

Adopted Levels

$Q(\beta^-) = -16110$ syst; $S(n) = 16830$ syst; $S(p) = -608$ 17; $Q(\alpha) = -8587$ 29 [2021Wa16](#)

$S(p), Q(\alpha)$: Deduced by the evaluator using mass excesses of -1487 17 for ^{34}K measured by [2024Dr01](#); -9384.3 4 for ^{33}Ar and 4675 24 for ^{30}Cl from [2021Wa16](#). Values from [2021Wa16](#): $S(p) = -880$ 200 (syst), $Q(\alpha) = -8320$ 200 (syst).

$\Delta Q(\beta^-) = 360$, $\Delta S(n) = 280$ (syst, [2021Wa16](#)).

$S(2p) = 2730$ 17, $Q(\varepsilon) = 16891$ 17, $Q(\varepsilon p) = 12227$ 17, from mass excesses of -1487 17 for ^{34}K measured by [2024Dr01](#); -13334.7 6 for ^{32}Cl , -18378.289 80 for ^{34}Ar , and -21003.3 4 for ^{33}Cl from [2021Wa16](#). Values from [2021Wa16](#): $S(2p) = 2460$ 200 (syst),

$Q(\varepsilon) = 17160$ 200 (syst), $Q(\varepsilon p) = 12490$ 200 (syst).

$S(2n) = 39350$ 450 (syst) ([2021Wa16](#)).

Isotope discovery: $\text{Be}(^{37}\text{Ca}, X) \rightarrow p + ^{33}\text{Ar}$ at NSCL, MSU ([2024Dr01](#)).

$S(2p)$ is positive and $S(p)$ is negative. ^{34}K is a ground state one-proton emitter.

Mass and separation energy calculations: [2021Ma33](#), [2022Zo01](#), [2019Zo02](#), [2013Ti01](#).

Other structure calculations: [2023Bo17](#), [2018Ka12](#).

 ^{34}K LevelsCross Reference (XREF) Flags

A $^9\text{Be}(^{37}\text{Ca}, X)$

E(level)	J^π	$T_{1/2}$	XREF	Comments
0	(1 ⁺)	<40 ns	A	$\%p = ?$ $T_{1/2}$: upper limit estimated from expected cross sections of a ^{40}Ca beam fragmentation on a Ni target at LISE, GANIL (1996PoZZ , 1986La17).
401 25	(2 ⁺)		A	
≈ 1240			A	
≈ 1810			A	